Reg. No. | | | | | | | | | |

Question Paper Code : 14104

B.E. / B.TECH DEGREE EXAMINATION, APRIL / MAY 2018

Third Semester

15MA304 - MATHEMATICS III – DISCRETE MATHEMATICS

Common to B.E. – Computer Science and Engineering & B.Tech. – Information Technology

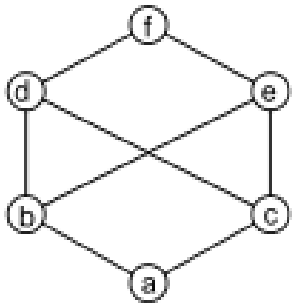
(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Let A = {1, 2, 3, 4}, and R is a relation defined by "a divides b". Write R as a set of ordered pair, Draw directed graph.
2. How many elements are there in A x A, A x B, A x φ and φ x φ where A has 'm' elements and B has 'n' elements.
3. Symbolize the following statement: "Neither Nehru nor Gandhi are naughty".
4. Produce the symbolic form of the statement " some animals are dangerous".
5. Draw a complete bipartite graph of $K_{2,3}$ and $K_{3,3}$.
6. Define a tree.
7. Define monoid with an example.
8. Define ring with an example.
9. Is the following Hasse diagram represent a lattice? Justify your answer.



10. Define Modular Lattice.

PART B – (5 × 16 = 80 Marks)

11. a) i. In a survey of 120 passengers, an airline found that 52 enjoyed wine with their meals, 75 enjoyed mixed drinks and 62 enjoyed iced tea. 35 enjoyed any given pair of these beverages and 20 passengers enjoyed all of them. Find the number of passengers who enjoyed

A. Only iced tea

B. Exactly two of the three beverages

C. None of the drinks. (8 Marks)

11. a) ii. Let f : R $\rightarrow$R and g : R$\rightarrow$R where R is the set of real numbers and f,g are given by f(x)=x^2-2, g(x) =x+4.

Verify "f" for bijective function and also find f og, gof . (8 Marks)

OR

11. b) i. Let P = { Straight lines} and

R= $\{(x, y) / x, y \in P$ and xRy iff x is parallel to y$\}$.

Verify whether R is an equivalence relation or not. (8 Marks)

11. b) ii. Let A={1,2,3,4} R ={(1,1),(1,2),(2,3),(2,4),(3,4),(4,1),(4,2} and S={ (3,1),(4,4),(2,3),(2,4),(1,1),(1,4)}. Find $R \circ R, S \circ R$.

Also find graph of $R \circ R$ and the matrix of $S \circ R$. (8 Marks)

12. a) i. Comment from the conclusion of the truth table to the expression $(p \leftrightarrow q) \vee (\neg p \leftrightarrow r)$ (8 Marks)

12. a) ii. Without using the truth table, show that the following argument is valid.

$L \rightarrow A, E \rightarrow \neg I, A \rightarrow E \Rightarrow I \rightarrow \neg L$ (8 Marks)

OR

12. b) i. Show that the following set of premises is inconsistent:

If the contract is valid, then John is liable for penalty. If John is liable for penalty, he will go bankrupt. If the bank will loan him money, he will not go bankrupt. As a matter of fact, the contract is valid and the bank will loan him money. (8 Marks)

12. b) ii. Verify the validity of the following inference:

"All humming birds are richly coloured". "No large birds live on honey". "Birds that do not live on honey are in colour". Therefore, "Humming birds are small" (8 Marks)

13. a) i. Prove that in a graph that number of odd degree vertices is even. (8 Marks)

13. a) ii. Use Dijkstra's algorithm to find the shortest pat between the vertices A and E in the following weighted matrix

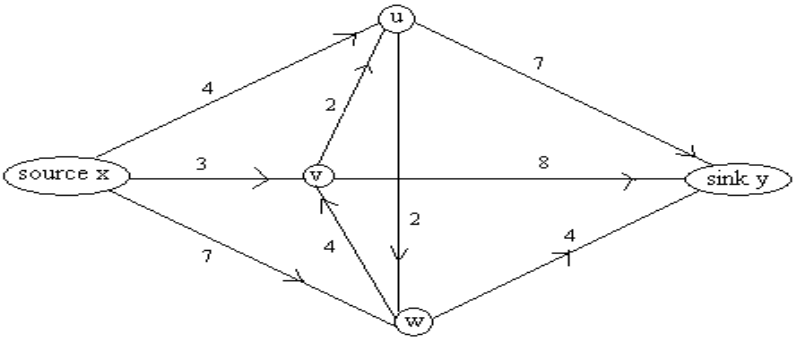
	A	B	C	D	E
A	0	1	2	0	0
B	1	0	1	3	2
C	2	1	0	1	2
D	0	3	1	0	4
E	0	2	2	4	0

(8 Marks)

OR

13. b) i. Explain Eulerien path, Eulerien circuit and Homiltonian path, Hmiltonium circuit and give an example to each. (8 Marks)

13. b) ii. Write down Kruskal's algorithm and also find the minimum spanning tree for the given weighted graph



(8 Marks)

14. a) i. Prove that the inverse element of a group is unique. (8 Marks)

14. a) ii. Let G be the group of all integers under addition and H be the group { 1,-1} under multiplication, define a map

f: (G,+) →(H,.) where $f(a) = \begin{cases} 1 & \text{, a is even} \\ -1 & \text{, a is odd} \end{cases}$, prove that

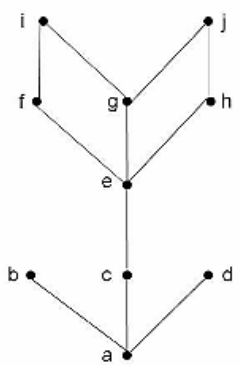
'f' is a homomorphism. (8 Marks)

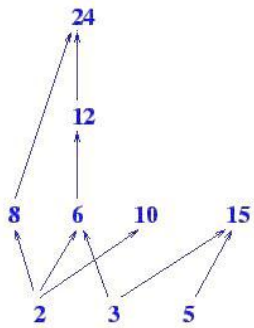
OR

14. b) i. State Lagrange's theorem and also find the left cosets of {[0], [3]} in the group $G = < Z_6, +6 >$. (8 Marks)

14. b) ii. Show that the set $A = \{1, \omega, \omega^2\}$ of all cubic roots of unity form an abelian group with respect to multiplication as composition. (8 Marks)

15. a) i. Calculate maximal, minimal, least, greatest element for the following Hasse diagrams.





(8 Marks)

15. a) ii. In a Boolean algebra if a+b=1 and a.b=0, show that the complement of every element is unique. (8 Marks)

OR

15. b) i. If P(S) is the power set of a set S and $\cup$ *and* $\cap$ are taken as join and meet, prove that $\{P(S), \subseteq\}$ is a lattice. (8 Marks)

15. b) ii. In any Boolean algebra , show that ab'+ba'=0 if and only if a=b. (8 Marks)